

Chapter 0: Introduction

**Computer Science: An Overview
Eleventh Edition**

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Computer science

- Computer science is the discipline that seeks to build a scientific foundation for such topics as computer design, computer programming, information processing, algorithmic solutions of problems, and the algorithmic process itself.
- It provides the underpinnings for today's computer applications as well as the foundations for tomorrow's computing infrastructure.

- This course provides a comprehensive introduction to this science.
- We will investigate a wide range of topics including most of those that constitute a typical university computer science curriculum.

Chapter 0: Introduction

- 0.1 The Role of Algorithms
- 0.2 The Origins of Computing Machines
- 0.3 The Science of Algorithms
- 0.4 Abstraction
- 0.5 An Outline of Our Study
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Terminology

- **Algorithm:** A set of steps that defines how a task is performed. For example, there are
- algorithms for cooking (called recipes),
- for finding your way through a strange city (more commonly called directions),
- for operating washing,
- For performing magic tricks

Terminology

- **Program:** A representation of an algorithm
- **Programming:** The process of developing a program
- **Software:** Programs and algorithms
- **Hardware:** Equipment

History of Algorithms

- The study of algorithms was originally a subject in mathematics.
- Early examples of algorithms
 - Long division algorithm
 - Euclidean Algorithm
- **Gödel's Incompleteness Theorem:** Some problems cannot be solved by algorithms.

Figure 0.2 The Euclidean algorithm

Description: This algorithm assumes that its input consists of two positive integers and proceeds to compute the greatest common divisor of these two values.

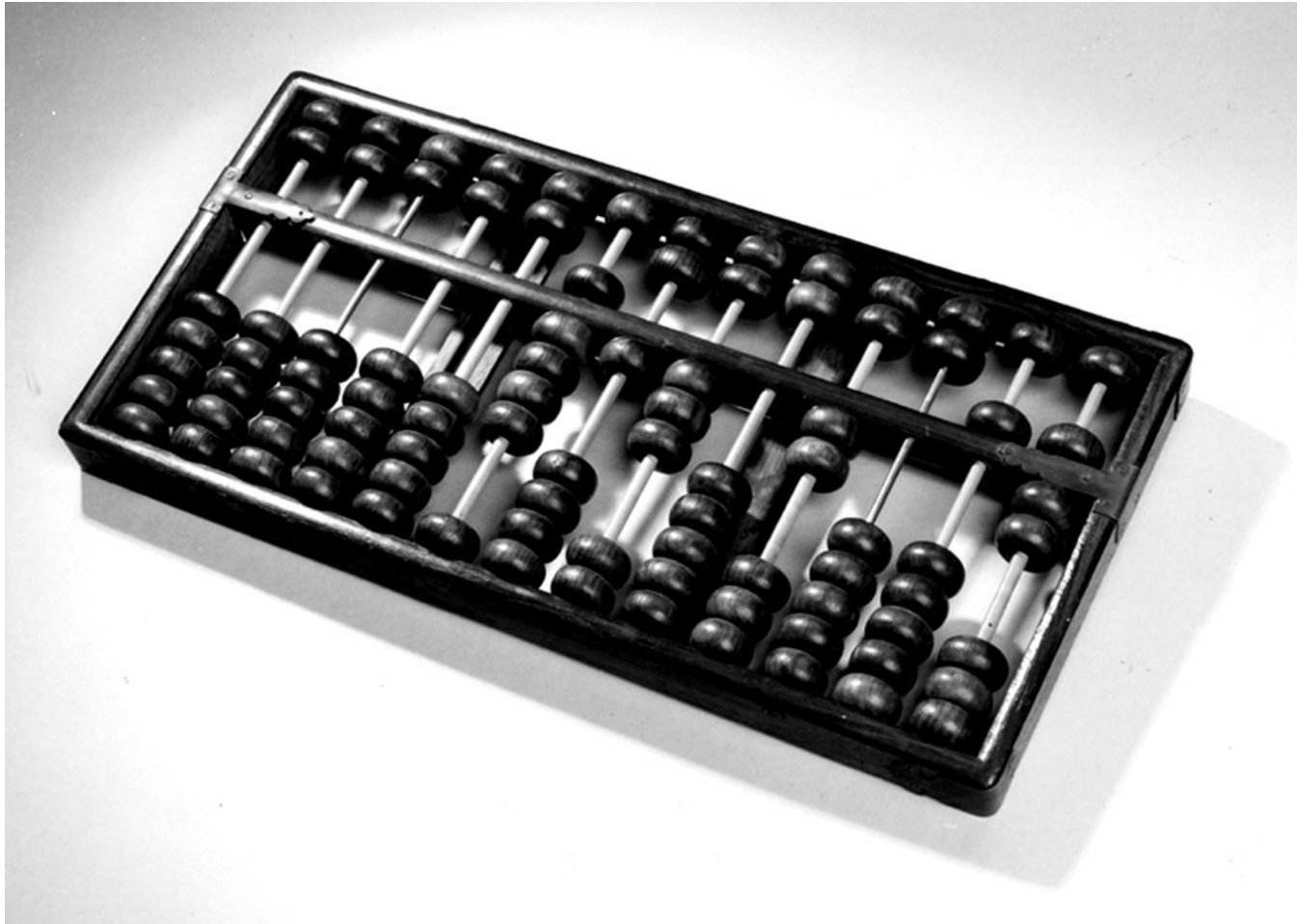
Procedure:

- Step 1. Assign M and N the value of the larger and smaller of the two input values, respectively.
- Step 2. Divide M by N, and call the remainder R.
- Step 3. If R is not 0, then assign M the value of N, assign N the value of R, and return to step 2; otherwise, the greatest common divisor is the value currently assigned to N.

Origins of Computing Machines

- Early computing devices
 - Abacus: positions of beads represent numbers
- As the beads are moved back and forth on the rods, their positions represent stored values. It is in the positions of the beads
- that this “computer” represents and stores data.
- Gear-based machines (1600s-1800s)
 - Positions of gears represent numbers
 - Blaise Pascal, Wilhelm Leibniz, Charles Babbage

Figure 0.3 An Abacus



Early Data Storage

- Punched cards
 - First used in Jacquard Loom (1801) to store patterns for weaving cloth
 - Storage of programs in Babbage's Analytical Engine
 - Popular through the 1970's
- Gear positions

Babbage's Difference Engine

The machines designed by Charles Babbage were truly the forerunners of modern computer design. If technology had been able to produce his machines in an economically feasible manner and if the data processing demands of commerce and government had been on the scale of today's requirements, Babbage's ideas could have led to a computer revolution in the 1800s. As it was, only a demonstration model of his Difference Engine was constructed in his lifetime. This machine determined numerical values by computing "successive differences." We can gain an insight to this technique by considering the problem of computing the squares of the integers. We begin with the knowledge that the square of 0 is 0, the square of 1 is 1, the square of 2 is 4, and the square of 3 is 9. With this, we can determine the square of 4 in the following manner (see the following diagram). We first compute the differences of the squares we already know: $1^2 - 0^2 = 1$, $2^2 - 1^2 = 3$, and $3^2 - 2^2 = 5$. Then we compute the differences of these results: $3 - 1 = 2$, and $5 - 3 = 2$. Note that these differences are both 2. Assuming that this consistency continues (mathematics can show that it does) we conclude that the difference between the value $(4^2 - 3^2)$ and the value $(3^2 - 2^2)$ must also be 2. Hence $(4^2 - 3^2)$ must be 2 greater than $(3^2 - 2^2)$, so $4^2 - 3^2 = 7$ and thus $4^2 = 3^2 + 7 = 16$. Now that we know the square of 4, we could continue our procedure to compute the square of 5 based on the values of 1^2 , 2^2 , 3^2 , and 4^2 . (Although a more in-depth discussion of successive differences is beyond the scope of our current study, students of calculus may wish to observe that the preceding example is based on the fact that the derivative of $y = x^2$ is a straight line with a slope of 2.)

x	x^2	First difference	Second difference
0	0		
1	1	1	
2	4	3	2
3	9	5	2
4	16	7	2
5			2

Early Data Storage

- Punched cards
 - Herman Hollerith (1860–1929), who applied the concept of representing information as holes in paper cards to speed up the tabulation process in the 1890 U.S. census.
 - Such cards ultimately came to be known as punched cards and survived as a popular means of communicating with computers well into the 1970s.

Early Computers

- The technology of the time was unable to produce the complex gear-driven machines of Pascal, Leibniz, and Babbage in a financially feasible manner. But with the advances in electronics in the early 1900s, this barrier was overcome.
- Based on mechanical relays
 - 1940: Stibitz at Bell Laboratories
 - 1944: Mark I: Howard Aiken and IBM at Harvard
- Based on vacuum tubes
 - 1937-1941: Atanasoff-Berry at Iowa State
 - 1940s: Colossus: secret German code-breaker
 - 1940s: ENIAC: Mauchly & Eckert at U. of Penn.

Figure 0.4 The Mark I computer



- the history of computing machines has been closely linked to advancing technology,
- including the invention of transistors (for which physicists William Shockley, John Bardeen, and Walter Brattain were awarded a Nobel Prize) and
- the subsequent development of complete circuits constructed as single units, called integrated circuits (for which Jack Kilby also won a Nobel Prize in physics).

- With these developments, the room-sized machines of the 1940s were reduced over the decades to the size of single cabinets.
- At the same time, the processing power of computing machines began to double every two years (a trend that has continued to this day).
- As work on integrated circuitry progressed, many of the circuits within a computer became readily available on the open market as integrated circuits encased in toy-sized blocks of plastic called chips.

Desktop computers

- A major step toward popularizing computing was the development of desktop computers.
- The origins of these machines can be traced to the computer hobbyists who built homemade computers from combinations of chips. It was within this “underground” of hobby activity that Steve Jobs and Stephen Wozniak built a commercially viable home computer and, in 1976, established Apple Computer, Inc. (now Apple Inc.) to manufacture and market their products.
- Other companies that marketed similar products were Commodore, Heathkit, and Radio Shack.

Personal Computers

- First used by hobbyists
- IBM introduced the PC in 1981.
 - Accepted by business
 - Became the standard hardware design for most desktop computers
 - Most PCs use software from Microsoft

Into the Millennia

- Internet revolutionized communications
 - World Wide Web
 - Search Engines (Google, Yahoo, and Microsoft)
- Miniaturization of computing machines
 - Embedded (GPS, in automobile engines)
 - Smartphone

Computer Science

- The science of algorithms
- Draws from other subjects, including
 - Mathematics
 - Engineering
 - Psychology
 - Business Administration

Computer Science

- researchers in different branches of computer science may have very distinct definitions of the science.
- For example, a researcher in the field of computer architecture may focus on the task of miniaturizing circuitry and thus view computer science as the advancement and application of technology.

Computer Science

- But, a researcher in the field of database systems may see computer science as seeking ways to make information systems more useful.
- And, a researcher in the field of artificial intelligence may regard computer science as the study of intelligence and intelligent behavior.

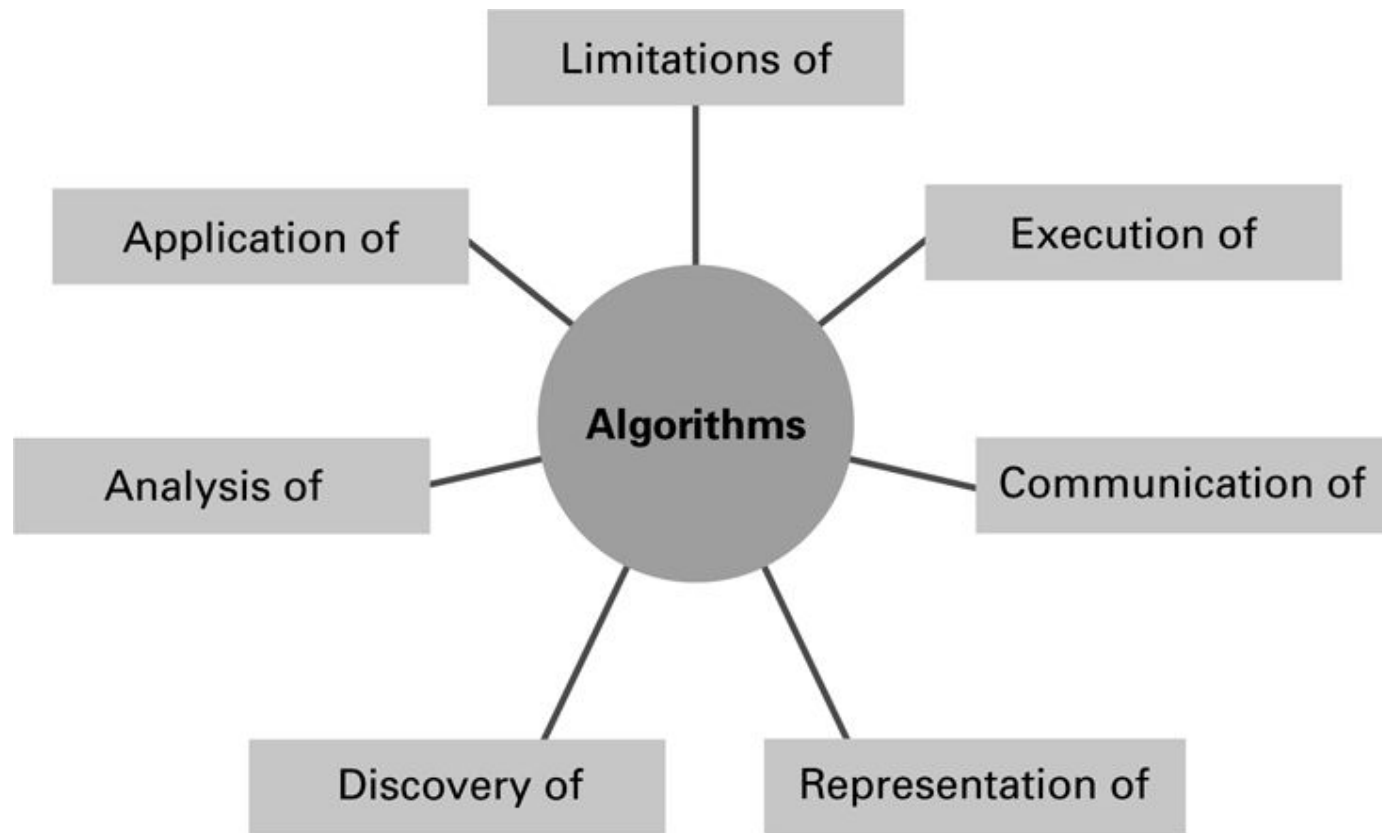
Central Questions of Computer Science

- Which problems can be solved by algorithmic processes?
- How can algorithm discovery be made easier?
- How can techniques of representing and communicating algorithms be improved?
- How can characteristics of different algorithms be analyzed and compared?

Central Questions of Computer Science (continued)

- How can algorithms be used to manipulate information?
- How can algorithms be applied to produce intelligent behavior?
- How does the application of algorithms affect society?

Figure 0.5 The central role of algorithms in computer science



Abstraction

- **Abstraction:** The distinction between the external properties of an entity and the details of the entity's internal composition
- It is abstraction that allows us to ignore the internal details of a complex device such as a computer, automobile, or microwave oven and use it as a single, comprehensible unit.

Abstraction

- Computers, automobiles, and microwave ovens are constructed from components, each of which is constructed from smaller components.
- Each component represents a level of abstraction at which the use of the component is isolated from the details of the component's internal composition.

Abstraction

- **Abstract tool:** A “component” that can be used without concern for the component’s internal properties
- This allows us to concentrate on how each component interacts with other components at the same level and how the collection as a whole forms a higher-level component.
- Thus we are able to comprehend the part of the system that is relevant to the task at hand rather than being lost in a sea of details.

Outline of Our Study

- Chapter 1: Data Storage
- Chapter 2: Data Manipulation
- Chapter 3: Operating Systems
- Chapter 4: Networks and the Internet
- Chapter 5: Algorithms
- Chapter 6: Programming Languages

Outline of Our Study (continued)

- Chapter 7: Software Engineering
- Chapter 8: Data Abstractions
- Chapter 9: Database Systems
- Chapter 10: Computer Graphics
- Chapter 11: Artificial Intelligence
- Chapter 12: Theory of Computation